

LINFENG WANG, PhD

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Skills

Language & Frameworks: Python, PyTorch, PyG, R, Bash, MySQL, HTML/CSS, C++
Machine learning: Scikit-learn, Transformers, LLMs, CNNs, RNNs, GNNs, VAEs, XGBoost
Development Tool: AWS, GCP, Docker, Git, Nextflow, Pandas, NumPy, Jupyter, Huggingface, LangChain, Numba
Visualization: Matplotlib, Plotly, ggplot2, streamlit, Flask

Experiences

Machiner learning engineer - Biographica, London UK

Apr 2026 –

- Develop, tested and benchmark **deep learning models** predicting gene expression from **DNA sequence and biological context**.
- Developed scalable **PyTorch training and fine-tuning pipelines** on **Azure and AWS infrastructure**, including dataset construction, model evaluation, hyperparameter optimisation and reproducible experimentation.
- Applied ML to **gene-editing and crop-trait prediction**, combining multi-omics data

PhD researcher - LSHTM (Clark Campino Phelan lab), London UK

Oct 2021 – Mar 2026

- Developed **dockerised** reproducible deep learning models (**CNN, RNN, GNN, Transformer**) with **PyTorch**
- Built **statistical pipelines** on **HPC** in Python for GWAS, PCA, logistic regression, and odds ratio analysis on **high dimensional data**
- Developed a **web application** and **backend software (PyPi package)** in **Python** for automating the design of amplicon primers.
- Developed an **RNN-based generative NLP model** for **novel antimicrobial peptide sequence design & classification with 91% accuracy**
- Developed an **GNN-based classification model** for natural selection prediction based on phylogenetic trees achieving **88% accuracy**
- Built and deployed a **RAG LLM** chatbot integrating **ChromaDB, LangChain**, and OpenAI API for course content and feedback collection

Machine learning consultant - Deep Science Venture, London UK

Mar 2025 – Oct 2025

Project: Sequence-Based Drug Discovery using Deep Generative Models

- Led **CNN, RNN, and VAE** models developments on biological sequences with data augmentation and hyperparameter optimization
- Producing a model of **86% accuracy** in predicting gene expression
- Applied **SHAP, LIME, and DeepLIFT** to extract interpretable biological insights from deep models
- Directed strategic model design for **generative sequence discovery** tasks

Data study group hackathon participant – The Alan Turing Institute, London UK.

Jan 2024

Project: Shallow Gas Hazard Detection in Offshore Seismic Data – [led to Turing publication see publication section](#)

- Led deep learning model development for detection and segmentation on geophysical imagery using **CNNs with contrastive learning**
- Delivered **93% classification accuracy** on legacy seismic datasets under 2-weeks
- Collaborated across domain boundaries to translate geoscience challenges **into ML solutions**

Machine learning Intern - Linkevity, London UK

Aug 2024 – Oct 2024

Project: Graph-Based Drug-Drug Interaction Prediction

- Recreated **GNN** models with with integration of **BERT** to predict drug compound interactions
- Trained **drug-drug interaction model** for **drug design, improved predictive performance by 15%**
- Scaled ML training workflows using **Google Cloud Platform** and **AWS** for high-throughput experimentation

Data Science intern - ByteDance, London UK

Aug 2022 – Feb 2023

Project: Neoantigen Insight Platform and Market Landscape Scouting

- Built **SQL-integrated chemical database** for drug candidates
- Used **autoencoders** to uncover hidden patterns in immune-related molecules across biological datasets
- Coordinated **interdisciplinary market research** on biologics and small molecules
- Authored bi-weekly **research briefings** and developed end-to-end ML pipelines for **drug discovery**.

SELECTED PUBLICATIONS (12 in total, 7 First authored expected)

- Wang et al. TOAST: A novel tool for designing targeted gene amplicons. *BMC genomics*. doi: <https://doi.org/10.1186/s12864-025-12247-9> - Website: <https://genomics.lshtm.ac.uk/webtoast/#/> - Pypi software package: <https://pypi.org/project/toast-amplicon/>
- Wang et al. LSTM-Based Models for Antimicrobial Peptide Classification and Generation. doi: <https://doi.org/10.1093/bioadv/vbaf274>
- Wang et al. Decoding Positive Selection with Phylogeny-Guided **Graph Attention Models**. doi: <https://link.springer.com/article/10.1186/s12859-026-06583-0>
- Wang et al. ML for stepwise prediction of TB treatment outcomes. preprint: <https://doi.org/10.21203/rs.3.rs-7558046/v1>
- DSG Report: British Geological Survey - Detecting Shallow Gas from Marine Seismic Images” The Alan Turing Institute (2025)
link: <https://www.turing.ac.uk/news/publications/data-study-group-final-report-british-geological-survey-0>

EDUCATION

UCL, London School of Hygiene and Tropical Medicine , DPhil Computational Genomics

Oct 2021 – Oct 2025

Dissertation: Integrative Genomic Sequencing and Machine Learning for TB Drug resistance, Diagnostic Tool Development

Award: UKRI – BBSRC – LiDo PhD scholarship

Imperial College London, MRes Bioengineering (Hons), Merit

Oct 2019 – Oct 2020

Relevant courses: *Data Analysis for Research*

King's College London, BSc Biochemistry (Hons), 1st

Sep 2016 – Jul 2019

Relevant courses: Bioinformatics, Protein design and functions, Molecular dynamic modelling

Awards: Wellcome Trust Biomedical Vacation Studentship (2018)